

Alexandra Catherine Montgomery-Whitfield

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San Francisco Bay Area, California, United States

<https://www.linkedin.com/in/alexandra-catherine-montgomery-whitfield-engineer>

Principal software engineer and engineering leader with 12+ years of experience architecting distributed systems, leading large teams, and shipping consumer and enterprise products at scale. Deep expertise across frontend, backend, infrastructure, and platform engineering. A pragmatic technologist who pairs long-term architectural vision with relentless execution. Reachable at <https://averylongportfoliourlthatdoesnotbreaknicely.example.com/portfolio/work/2024> for a full case-study archive.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Rust, Java, Kotlin, Ruby, C++, SQL

Frontend: React, Next.js, Vue, Svelte, Tailwind CSS, Redux, React Query, Webpack, Vite, Storybook

Backend: Node.js, Express, NestJS, GraphQL, gRPC, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ

Cloud and DevOps: AWS, GCP, Azure, Docker, Kubernetes, Terraform, Pulumi, GitHub Actions, Prometheus, Grafana

Practices: Distributed Systems Architecture and Design, Event-Driven Microservices, Domain-Driven Design, Observability, Site Reliability Engineering

Leadership: Team Building, Mentorship, Technical Strategy, Cross-functional Collaboration, Hiring and Interviewing

PROFESSIONAL EXPERIENCE

Principal Software Engineer and Technical Lead

2021-06 - Present

Hyperscale Cloud Infrastructure Corporation

San Francisco, CA

- Achievement 1: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 2: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 3: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.
- Achievement 5: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 6: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.
- Achievement 7: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

Staff Software Engineer

2019-01 - 2021-05

Meridian Financial Technologies

New York, NY

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- Achievement 6: shipped, measured, and iterated on a meaningful improvement that moved a key business metric and was adopted across the wider organisation.

Senior Full-Stack Engineer

2016-08 - 2018-12

Nimbus Analytics

Remote

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Software Engineer II

2014-03 - 2016-07

BrightPath Software Solutions and Consulting Group International

Austin, TX

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Founding Engineer

2012-09 - 2014-02

StartupForge

Palo Alto, CA

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Junior Developer (no achievements listed)

2011-06 - 2012-08

Legacy Systems Inc.

Boston, MA

Engineering Intern

2010-06 - 2010-09

Internship Co.

Seattle, WA

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SELECTED PROJECTS

Open-Source Distributed Job Scheduler

<https://github.com/alexandra-mw/distributed-scheduler>

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Real-Time Collaborative Document Engine with CRDT Conflict Resolution

<https://crdt-docs.example.com>

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Project Without Any Description (edge case)

<https://no-description.example.com>

Personal Site Generator

<https://github.com/alexandra-mw/site-gen>

- A small static-site generator written in Go with incremental builds and a plugin system.

EDUCATION

Massachusetts Institute of Technology

2009 - 2011

Master of Science in Computer Science and Artificial Intelligence

Cambridge, MA

- Thesis on fault-tolerant consensus in geo-distributed databases
- Graduate Teaching Assistant for Distributed Systems
- GPA 3.9 / 4.0

University of California, Berkeley

2005 - 2009

Bachelor of Science in Electrical Engineering and Computer Science

Berkeley, CA

- Graduated summa cum laude
- President of the ACM student chapter

Online Continuing Education (no highlights)

2020 - 2020

Professional Certificate in Machine Learning

CERTIFICATIONS AND AWARDS

- AWS Certified Solutions Architect - Professional
- Google Cloud Professional Cloud Architect
- Certified Kubernetes Administrator (CKA)
- Certified Kubernetes Application Developer (CKAD)
- HashiCorp Certified: Terraform Associate
- Microsoft Certified: Azure Solutions Architect Expert

- Distinguished Engineering Award, 2023
- Hackathon Grand Prize Winner, Global Cloud Summit, 2022

SPEAKING AND PUBLICATIONS

- Keynote: Designing Resilient Systems at Planet Scale - Global Cloud Summit 2023.
- Conference talk: The Pragmatic Path to Microservices - QCon San Francisco 2022.
- Co-author of a widely cited paper on consensus protocols in geo-distributed databases.
- Regular contributor to several popular open-source infrastructure projects.
- Led a cross-functional initiative spanning multiple teams and time zones, coordinating engineering, design, and product stakeholders to deliver a high-availability platform serving millions of monthly active users while reducing infrastructure cost by a measurable margin and improving p95 latency across critical paths.

LANGUAGES

- English (Native)
- Spanish (Fluent)
- French (Professional)
- German (Conversational)
- Mandarin (Basic)

ADDITIONAL LINKS OR DATA

Portfolio: <https://alexandra.dev>

GitHub: <https://github.com/alexandra-mw>

Case Studies: <https://averylongportfoliourlthatdoesnotbreaknicely.example.com/case-studies>

X / Twitter: @alexandra_mw